

Review

Time as a danger signal promoting G1 arrest after mitosis

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Cell cycle checkpoints preventing the replication and inheritance of damaged DNA are crucial for maintaining genome stability and stopping the growth of damaged cells. Canonical checkpoints do this by preventing passage between cell cycle phases until damage has been repaired, or by promoting cell cycle exit. Herein we review checkpoint integration between cell cycle phases, specifically findings showing that extended spindle assembly checkpoint surveillance in mitosis is a danger signal triggering G1 cell cycle arrest. Evidence linking mitotic delays induced by activation of the spindle assembly checkpoint with positive and negative regulators of the G1 DNA damage checkpoint target p53 is discussed, with a focus on time-dependent changes to a p53-binding deubiquitinating complex USP28–53BP1 and the p53 ubiquitin-ligase mouse double minute homologue 2 (MDM2), respectively.

A brief history of time in mitosis

Cell cycle regulation by conserved checkpoint pathways is necessary for maintaining homeostasis at both the cellular and organismal level (Box 1). Loss of this regulation leads to unregulated cell growth and proliferation, manifesting in pathological states such as genome instability and cancer [1,2]. Although inactivation of cell cycle checkpoints is commonly considered in the context of cancer, aberrant activation of cell cycle checkpoints can lead to developmental disorders and is associated with loss of proliferation during ageing [3,4]. Because of these associations, much of what we know about cell cycle checkpoint mechanisms has been gleaned from research conducted in already genomically unstable cancer cell lines. In such cell lines, one or more of these checkpoints is often inactivated or altered by mutation or other means, including epigenetic changes and antagonism through viral oncogenes [5–7]. Understanding how checkpoints operate under normal conditions to protect the integrity of the cell cycle and safeguard against genome instability is therefore of great importance. While often viewed as independent pathways, it has become evident that canonical cell cycle checkpoints do not normally work in isolation within a single cell cycle phase and can influence events in subsequent phases. This type of checkpoint integration was observed in a landmark study showing that nontransformed diploid cells arrest in G1 phase if the preceding mitosis is transiently prolonged due to activation of the **spindle assembly checkpoint (SAC) (see Glossary) (Box 2)** with mitotic spindle poisons [8]. A key breakthrough in this work was the demonstration that the **Gap1 phase (G1)** arrest following a mitotic delay exceeding a threshold time of approximately 90 min was dependent on the activity of the transcription factor **p53** [8]. Although p53 is often considered in the context of the G1 DNA damage checkpoint (Box 3), DNA damage was not detected in arrested cells, and moreover the arrest was independent of canonical DNA damage kinases [8]. This suggested that the cause of the p53-dependent G1 arrest was a consequence of the increased time spent in mitosis and not DNA damage per se [8]. Further support for this interpretation came from studies showing that mitotic delays also explain the G1 cell cycle arrest seen after chromosome mis-segregation in aneuploid cells [9], and after delays in chromosome alignment caused by **centromere-**

Highlights

Time is a danger signal directing potentially damaged mitotic cells to p53-dependent G1 arrest.

Mouse double minute homologue 2 (MDM2) functions as a molecular timer monitoring the length of mitosis.

If MDM2 drops below a threshold concentration at the end of mitosis, p53 is stabilised in G1 daughter cells, triggering cell cycle arrest through p21 induction.

USP28–53BP1 stopwatch complexes reinforce p53-dependent G1 arrest, potentially by antagonising MDM2 ubiquitination of p53.

Loss of the mitotic timer pathway enables proliferation of damaged or otherwise genomically unstable cells.

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Box 1. Canonical cell cycle checkpoints

To create new cells from existing cells the genome must be replicated to completion and subsequently divided equally between two daughter cells. This cyclical process termed the cell cycle (Figure 1A) begins in G1 phase, where the cell grows and prepares for DNA replication in **S phase**. Following DNA replication, cells enter another growth phase termed G2 where they prepare for cell division in the succeeding **mitotic phase (M phase)**, where both the genome and cytoplasm are divided equally to form two daughter cells. To facilitate regulated progression through these crucial phases of the cell cycle, quality control mechanisms (termed checkpoints) are employed to govern transitions between these stages. These protective checkpoints are in place to limit passage into the next cell cycle phase if a problem is detected. For example, if DNA damage occurs in G1 phase, cells arrest in G1 until the damage is repaired to prevent the replication of damaged DNA [70,71]. If DNA damage occurs in G2 phase, the G2 checkpoint arrests cells in G2 until the damage is resolved, to limit the inheritance of damaged chromosomes during M phase [72,73]. In mitosis, the SAC prevents anaphase onset until all chromosomes are captured and correctly aligned along the metaphase plate [21] (Box 2). Collectively, these checkpoints work together to protect against genome instability and aneuploidy. In tissues or during development, different genotoxic stresses, including delays in mitosis, promote G1 arrest and withdrawal from the proliferative cell cycle (Figure 1B). This protective response limits proliferation of potentially damaged cells and protects multicellular organisms from genome instability and associated disease states.

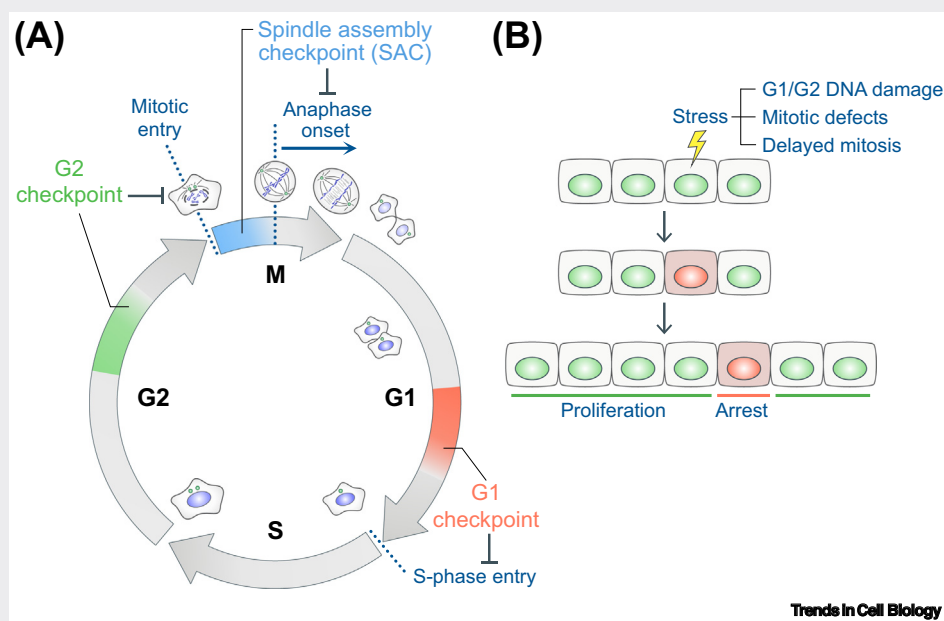


Figure 1. The cell cycle and its associated checkpoints.

associated protein E (CENP-E) kinesin inhibition [10]. Crucially, in the absence of any drug treatments or genetic changes, stochastic variation in mitotic duration due to the inherent properties of chromosome alignment and error correction pathways also causes G1 cell cycle arrest when mitosis exceeds a time threshold of 60–90 min in the absence of any detectable DNA damage [11]. It is important to highlight that chromosome mis-segregation per se does not necessarily lead to cell cycle arrest in the ensuing G1 daughter cells. Indeed, artificially forcing chromosome mis-segregation by inhibition of the SAC triggers premature mitotic exit, and cells therefore remain below the threshold time in mitosis required for G1 arrest, despite being visually aneuploid [10,12]. Furthermore, aneuploidy does not appear to be directly detected since some whole chromosome numerical aneuploidies resulting from SAC failure can be propagated in p53 wild-type cells [10]. By contrast, structural aneuploidies resulting from aberrant DNA damage and repair cannot be propagated in p53 wild-type cells, possibly due to ongoing DNA damage signalling [10]. In addition to mitotic delays, p53 has also been implicated in the response to other types of mitotic perturbation,

Glossary

Anaphase-promoting complex/cyclosome (APC/C): the major ubiquitin-ligase functioning during M phase.

ATM: ataxia-telangiectasia mutated; a major upstream kinase in the DNA damage response.

BUB3 (budding uninhibited by benzimidazoles 3): a serine/threonine protein kinase functioning as part of the MCC.

BubR1 (Bub1-related 1): a regulatory component of the MCC.

Caspase-2: a cysteine protease triggering apoptotic cell death after genotoxic stress; a key enzyme in the apoptotic response.

Cell division cycle 20 (CDC20): a coactivator of the APC/C.

Centromere-associated protein E (CENP-E): a kinesin involved in chromosome alignment and segregation.

Cereblon (CRBN): the substrate recognition subunit of the Cullin 4A ubiquitin-ligase complex.

Checkpoint kinase 2 (Chk2): a key kinase in the DNA damage response acting downstream of ATM.

Chromosome passenger complex (CPC): the regulatory component controlling Aurora kinase B localisation during mitosis.

Clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated protein 9 (Cas9): a technique for gene editing.

Cyclin-dependent kinase (CDK): a key enzyme that regulates passage through the cell cycle.

Enzyme 3 (E3): the ubiquitin ligase in the ubiquitination reaction.

G0: nonproliferative or quiescent phase at the start of the cell cycle.

Gap1 phase (G1): the first growth phase of the cell cycle.

Gap2 phase (G2): the second growth phase of the cell cycle.

Kinetochores scaffold 1 (KNL1): a key signalling scaffold for the SAC and MCC assembly.

Mitotic arrest-deficient protein 2 (MAD2): a regulatory component of the MCC.

Mitotic checkpoint complex (MCC): the functional effector of the SAC that prevents APC/C activation until all chromosomes are correctly aligned and bioriented.

including cytokinesis failure and supernumerary centrosomes, which can promote chromosome mis-segregation and cause numerical or structural aneuploidy [13–19]. However, in such cases it is often challenging to determine whether DNA damage or delayed mitosis is the cause of the p53-dependent cell cycle arrest. Collectively, these studies suggested that a key factor being monitored and responded to in aneuploid cells is the increased duration of mitosis, and that in some cases this may be accompanied by DNA damage.

This review focuses on the idea that time is a danger signal, and on our current understanding of how mitotic delays are detected and responded to at the molecular level through integration of events in mitosis with response pathways in G1. Finally, the role of these important safeguarding pathways in disease states – including cancer and developmental disorders – is discussed, alongside future therapeutic opportunities and research challenges.

Time as a danger signal: a conceptual framework for the mitotic timer pathway

As anyone who has ever waited for a flight, train, or bus knows, unplanned delays are both stressful and a sign of problems in the system that generally indicate something is wrong. Delays in mitosis are most simply understood as an indicator that the cell has encountered a problem in aligning and segregating its chromosomes. While there could be many possible reasons why mitosis is delayed, including environmental perturbations or mutations, the common denominator is that errors in mitosis generally present as an increase in the period of SAC activation. This lengthening of mitosis is often considered a beneficial outcome since the SAC helps reduce errors in chromosome segregation [20,21]. However, this comes at the cost of benefiting aneuploid cancer cells or otherwise damaged cells which often take longer to align and segregate their chromosomes [22,23]. Thus, it is advantageous for multicellular organisms to possess a mechanism that could monitor delays in mitosis and relay a danger signal preventing proliferation of these potentially damaged and genomically unstable cells. The mitotic timer pathway is proposed to serve this purpose. Accordingly, when mitosis is delayed beyond a threshold time, the mitotic timer mechanism channels these potentially damaged cells into p53-dependent cell cycle arrest in the ensuing G1 to limit

Box 2. Spindle assembly checkpoint

During mitosis, spindle microtubules emanating from the spindle poles capture and orientate the chromosomes in prometaphase to enable faithful chromosome segregation following the metaphase-to-anaphase transition (Figure 1) [20,21]. Biorientation of the chromosomes on the mitotic spindle is a prerequisite for progression through the metaphase-to-anaphase transition, and chromosome segregation is prevented until all chromosomes have been correctly captured and aligned along the metaphase plate. This delay in mitotic progression is achieved by the spindle assembly checkpoint (SAC), which monitors and corrects aberrant or missing kinetochore–microtubule attachments and prevents chromosome segregation until all such errors have been eliminated. These processes depend on two serine/threonine kinases, Aurora B within the **chromosomal passenger complex (CPC)** at inner centromeres, and the mitotic checkpoint kinase **monopolar spindle 1 (MPS1)**, which is dynamically recruited to kinetochores [21]. The Aurora B error correction pathway detects and resolves incorrect microtubule–kinetochore attachments via phosphorylation of kinetochore components, and in addition promotes recruitment of MPS1 which also helps to resolve erroneous attachments. At these checkpoint active kinetochores, MPS1 phosphorylates **kinetochore scaffold 1 (KNL1)** to initiate a cascade that drives assembly of the **mitotic checkpoint complex (MCC)** [74]. The MCC, composed of **mitotic arrest-deficient protein 2 (MAD2)**, **budding uninhibited by benzimidazole-related 1 (BubR1)**, **budding uninhibited by benzimidazoles 3 (BUB3)**, and **cell division cycle 20 (CDC20)**, is the critical effector of the SAC which binds to and inhibits the CDC20-bound form of the anaphase-promoting complex/cyclosome (APC/C), an E3 ligase which ubiquitinates key substrates, including cyclin B and securin, to trigger anaphase [75]. Thus, whilst the SAC remains active, MCC binding and inhibition of APC–CDC20 prevents the metaphase-to-anaphase transition. Once all chromosomes are correctly aligned on the mitotic spindle the SAC is turned off, MCC production ceases, releasing active APC–CDC20 which triggers cyclin B and securin destruction, promoting anaphase onset and mitotic exit. Microtubule poisons [nocodazole, paclitaxel (Taxol)] and kinesin motor protein inhibitors (Eg5 inhibitors, CENP-E inhibitors) – activate the SAC and delay anaphase onset until the chemicals are removed. Additionally, cells with aneuploid or tetraploid genomes have been shown to be more dependent on the SAC for survival, presumably because the elevated number of chromosomes take longer to align [22,23].

Mitotic phase (M phase): a phase involving mitosis and cytokinesis, the division of the genome and cytoplasm, respectively.

Monopolar spindle 1 (MPS1): the key kinase regulator of the SAC.

Mouse double minute homologue 2 (MDM2): the ubiquitin ligase for p53.

p21: protein of 21 kDa, a transcriptional target of p53 that directly inhibits CDK complexes.

p53: protein of 53 kDa, a key transcription factor activated in response to stress.

p53 binding protein 1 (53BP1): p53 binding protein 1; a scaffold protein with roles in DNA damage repair and mitotic timer arrests.

PIDDosome: p53-induced protein with a death domain 1 (PIDD1)-containing complex; an activation platform for caspase-2, proposed to trigger MDM2 cleavage under certain stresses.

Polo-like kinase 1 (PLK1): a kinase regulating mitotic entry, progression, and exit.

Polo-like kinase 4 (PLK4): a kinase regulating centriole biogenesis.

PROteolysis-Targeting Chimera (PROTAC): bivalent degrader drugs acting to destroy the protein of interest.

Protein phosphatase magnesium-dependent 1D (PPM1D): a phosphatase functioning in the DNA damage response; also known as WIP1.

S phase: synthesis phase, the cell cycle phase in which DNA replication occurs.

Short-hairpin RNA (shRNA): an artificial RNA molecule used for gene silencing through RNA interference.

Spindle assembly checkpoint

(SAC): the key mitotic surveillance checkpoint delaying anaphase onset until all chromosomes are correctly captured and aligned.

Tripartite motif-containing protein 37 (TRIM37): a ubiquitin-ligase regulating centrosome function.

Ubiquitin-specific protease 28 (USP28): a deubiquitinase reported to act on p53.

WIP1: wild-type p53-induced phosphatase 1 (also known as PPM1D), a phosphatase functioning in the DNA damage response.

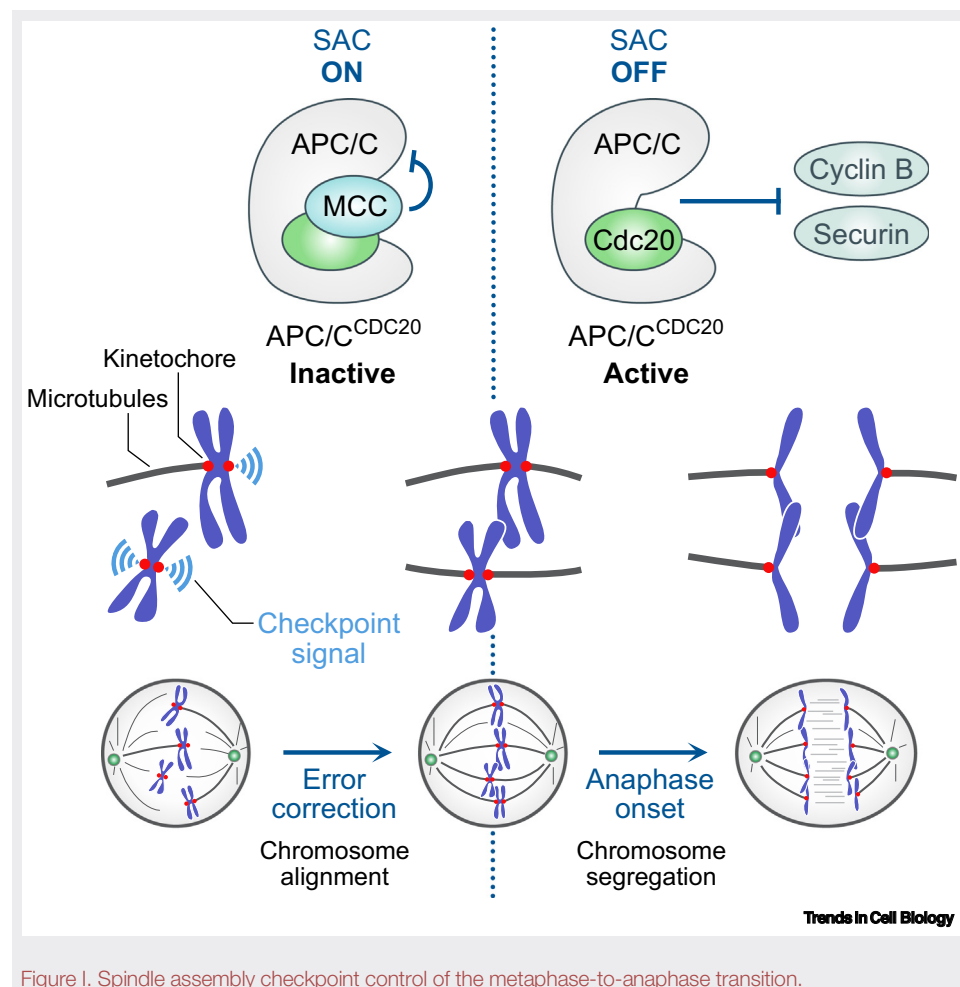


Figure 1. Spindle assembly checkpoint control of the metaphase-to-anaphase transition.

their further proliferation (Figure 1A). Identification of the molecular components underpinning this behaviour have therefore been the subject of a growing body of research.

How can delays in mitosis beyond the period required for segregation of a normal diploid genome trigger a p53-dependent cell cycle arrest in the following G1 phase? Conceptually, there are two simple possibilities: either time-dependent loss, or accumulation of an activity inhibiting or promoting G1 cell cycle arrest, respectively (Figure 1B). Emerging evidence provides support for both proposals, and suggests that these distinct mechanisms reinforce one another to converge on the key G1 checkpoint protein p53.

Discovery of a mitotic timer: MDM2

In the absence of stress, p53 is maintained at a low concentration in G1 and other cell cycle phases due to ongoing ubiquitination by its **enzyme 3 (E3)** ubiquitin-ligase **mouse double minute homologue 2 (MDM2)**, leading to proteasomal destruction (Figure 2A) [24–26]. In addition to regulation of p53, MDM2 is also capable of ubiquitinating itself and hence has a half-life of <30 min in most cell lines tested [11,27]. Therefore, MDM2 requires ongoing protein synthesis to balance destruction and maintain its steady state concentration. These properties of MDM2 have

Box 3. Canonical pathway for regulation of p53

The transcription factor and tumour suppressor protein p53 acts as the central regulator of the cellular stress response, integrating and in turn responding to a variety of different cellular stress signals (Figure 1) [70,76]. Under resting conditions, the basal level of p53 is kept low by the opposing ubiquitin-ligase MDM2, which promotes the ubiquitination and subsequent proteasomal degradation of p53 [11,24–26]. However, in response to cellular stress such as DNA damage, a cascade of associated kinases phosphorylates p53 and MDM2, disrupting their interaction and suppressing p53 ubiquitination [77]. This leads to stabilisation and thus accumulation of p53, which facilitates the formation of a tetrameric, transcriptionally active complex. The active transcription factor can then upregulate a whole host of genes controlling DNA repair, apoptosis, and cell cycle arrest [78]. In terms of cell cycle arrest, the CDK inhibitor protein p21 is the key p53 target [79,80]. Importantly, the p53 antagonist MDM2 is also a transcriptional target of p53, thereby allowing p53 to regulate its own stability through a negative feedback loop [81]. Thus, when p53 levels fall, MDM2 levels decline too, allowing p53 to rise again. This tightly regulated feedback system operates to maintain p53 concentration in a balanced steady state, promoting continued proliferation in the absence of stress, whilst retaining the potential for rapid and enforced p53 upregulation in response to damaging stimuli [75]. Ultimately, this prevents uncontrolled proliferation and maintains genomic integrity.

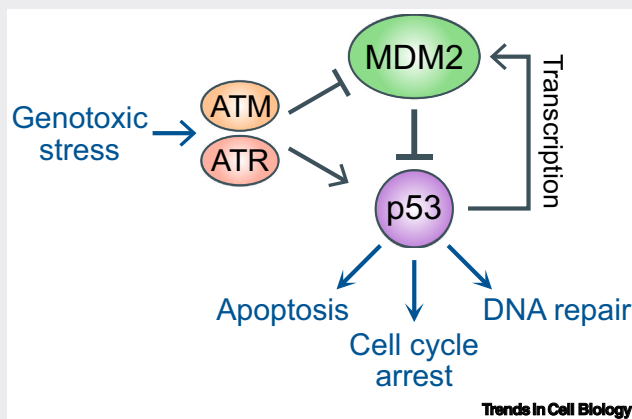


Figure 1. The p53-dependent cellular stress response.

important consequences when cells enter mitosis, a period of the cell cycle where transcription and translation are both largely attenuated [28,29]. By contrast, protein destruction by the ubiquitin–proteasome system continues in M phase [30]. Thus, when cells enter mitosis, MDM2 levels start to fall due to ongoing self-catalysed destruction and attenuation of protein synthesis, whereas p53 levels remain stable or rise slightly [11]. Furthermore, because MDM2 mRNA is destroyed during mitosis, re-accumulation of MDM2 protein in G1 daughter cells is dependent on *de novo* p53-dependent transcription of MDM2 [11,28]. These properties are consistent with proposals that MDM2 possesses the characteristics predicted for a mitotic timer protein [1,31]. By contrast, p53 mRNA remains stable during mitotic delays, and hence G1 daughter cells inherit p53 transcripts as well as the p53 protein [11,28]. After a prolonged mitosis, this creates a window of opportunity in early G1 in which p53 can trigger cell cycle arrest before MDM2 recovers to its normal levels following a lag period [11]. Direct experimental tests of this hypothesis in diploid cells with a normal centrosome complement show that mitosis typically completes in under 60 min, leaving sufficient MDM2 to limit p53 stability and activity in G1 daughter cells [11]. When mitosis is delayed beyond a threshold time between 60 and 90 min, there is insufficient MDM2 in early G1 to oppose p53-dependent induction of the **cyclin-dependent kinase (CDK) inhibitor p21** and subsequent G1 cell cycle arrest (Figure 2B) [11].

If MDM2 is the critical time-sensitive component, then lowering the concentration of MDM2 in mitosis in the absence of mitotic delays, or increasing MDM2 concentration in cells with mitotic delays, should be sufficient to trigger or suppress cell cycle arrest in G1, respectively. These

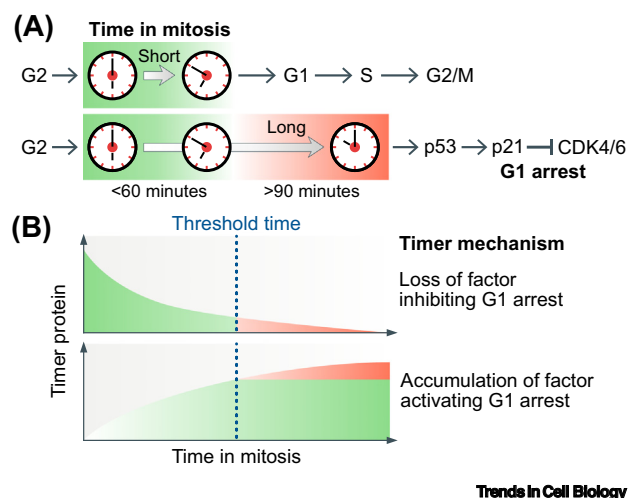


Figure 1. A conceptual framework for the mitotic timer pathway. (A) Delays in mitosis beyond a threshold time of 60–90 min result in p53-dependent G1 arrest in the ensuing G1 daughter cells. Mechanistically, once stabilised and activated, p53 induces the upregulation of its target gene, the cyclin-dependent kinase (CDK) inhibitor protein p21. Through directly inhibiting CDK4/6 G1 complexes, p21 maintains the G1 arrest. (B) Conceptually, p53 activation following mitotic delays could be due to either loss of a factor inhibiting G1 arrest, or accumulation of a factor activating G1. The balance of these opposing activities helps set the threshold for G1 cell cycle arrest following a prolonged mitosis.

predictions were directly tested by triggering MDM2 destruction in a normal length (<60 min) mitosis using a specific MDM2-directed **PROteolysis-Targeting Chimera (PROTAC)** and with cell lines engineered to overexpress MDM2. The PROTAC MD-224 is a bivalent drug that triggers rapid and selective MDM2 destruction by the **cereblon (CRBN)** E3 ubiquitin ligase and the proteasome regardless of the cell cycle phase [11,32]. Cells depleted of MDM2 using MD-224 during a normal length mitosis responded as if more time had passed, and initiated a robust p53-dependent induction of p21 and sustained cell cycle arrest of the G1 daughter cells [11]. Conversely, cells engineered to express fivefold higher levels of MDM2 behaved as if less time had passed and showed greatly reduced G1 arrest after 4 h mitotic delays, a condition that triggers G1 arrest in cells with normal levels of MDM2 [11]. Thus, the concentration of MDM2 remaining at the end of mitosis and start of G1 is a key determinant of whether cells are channelled into a cell cycle arrest. A further element of the proposed timer mechanism is that MDM2 destruction is not regulated by the **anaphase-promoting complex/cyclosome (APC/C)** ubiquitin ligase which is the core regulator of mitotic exit in every cell cycle [11]. MDM2 destruction thus proceeds inexorably during SAC-dependent mitotic arrests when the APC/C is inhibited. Consistent with this view, MDM2 also lacks an APC/C degron. Thus, the autonomous nature of MDM2 destruction affords cells a simple way of measuring the duration of mitosis, whereby the drop in MDM2 concentration is then read out by p53 in G1. Because this mechanism can be uncoupled from time in mitosis by directly manipulating MDM2 concentration with the MD-224 PROTAC within a single cell cycle, it is reasonable to conclude that loss of MDM2 is the primary time-sensing component associated with delayed progression through mitosis.

A mitotic stopwatch complex: USP28–53BP1

Alterations in the number of centrioles and centrosomes, including their loss, have long been known to correlate with increased frequency of chromosome instability in cells isolated from aneuploid tumours [33]. Following the discovery that **polo-like kinase 4 (PLK4)** is the primary regulator of centriole biogenesis [34,35] and the development of chemical inhibitors of PLK4 [36], research into the function of centrosomes has converged with studies of the mitotic timer. Although it was initially reported that PLK4 inhibition with the drug centrinone led to reversible centriole loss without extending mitotic length [36], subsequent observations revealed that PLK4 inhibition in diploid human cells with normal centriole copy number results in progressive mitotic delays over multiple cell cycles, until the resulting centrosome-depleted cells arrest in G1 [37–39].

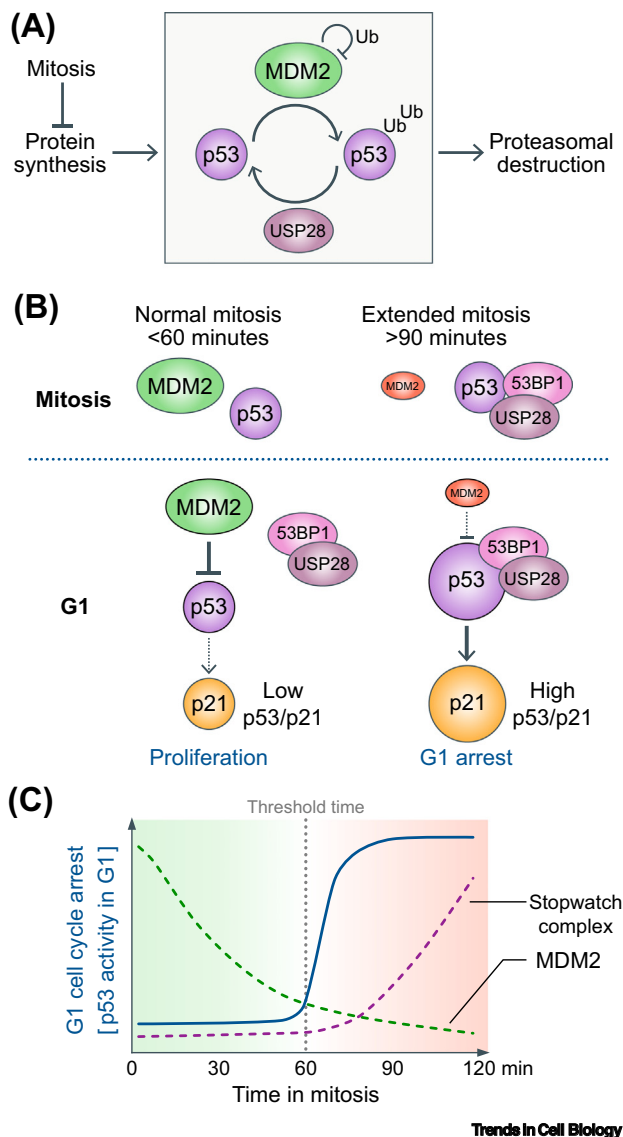


Figure 2. Mechanistic details of the mitotic timer pathway. (A) The cellular levels of p53 are maintained through regulation of p53 stability. Mouse double minute homologue 2 (MDM2) targets p53 for ubiquitination, marking p53 for proteasomal destruction. The p53 deubiquitinating enzyme USP28 may antagonise this process. (B) Following a normal length mitosis, there is sufficient MDM2 present in G1 daughter cells to limit p53 activation. Cells do not arrest in G1 but retain capacity to arrest in response to signals such as G1 DNA damage (left). Following a prolonged mitosis, MDM2 is depleted during the mitotic delay and p53 is stabilised in G1 daughter cells leading to p21 induction and G1 cell cycle arrest. USP28–53BP1 stopwatch complexes may help reinforce p53 stabilisation in the absence of the antagonistic MDM2 activity (right). (C) A model for p53 activation following a mitotic delay. During mitotic delays MDM2 concentration falls below a threshold, affording p53 an unopposed window to rise in G1 daughter cells. USP28–p53BP1 stopwatch complexes rise in activity during prolonged mitoses and, in the absence of MDM2, help set the p53 activation threshold required for stable G1 arrest.

Thus, centrosome depletion indirectly leads to p53-dependent G1 arrest due to an increased time spent in mitosis caused by slowed assembly of the mitotic spindle and chromosome alignment, a conclusion supported by observations showing progressive increases in mitotic duration as cells lose centrioles [40].

To better understand the mechanism of the response to PLK4 inhibition, the same studies focused on the identification of components that would, when deleted using **clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated protein 9 (Cas9)** gene editing, allow centrosome-depleted cells to continue proliferating and bypass the G1 arrest. In addition to the expected candidates p53 and its downstream effector protein p21 (Box 3), all three studies identified the DNA damage scaffolding protein **p53 binding protein 1 (53BP1)** and the deubiquitinase **ubiquitin-specific protease 28 (USP28)** as key factors required for G1 arrest following the removal of centrosomes [37–39]. One screen for centrinone resistance

also identified **tripartite motif-containing protein 37 (TRIM37)**, a ubiquitin ligase and known negative regulator of centrosome function [39,41]. Knockout of TRIM37 promotes spindle formation and reduces mitotic delays in acentrosomal cells created by PLK4 inhibition and thus suppresses G1 arrest by the timer pathway [42,43]. These findings support the conclusion that it is prolonged mitosis upon loss of centrosomes that triggers activation of p53 and the mitotic timer pathway. Despite these findings, TRIM37 is not considered a core mitotic timer component since it has no impact on G1 arrest caused by mitotic spindle poisons. Unlike TRIM37, USP28 and 53BP1 were found to be required for G1 arrest in response to both loss of centrosomes and mitotic delays created with spindle poisons [38,39]. One thing to note is that essential negative regulators of p53, such as MDM2, which lead to p53 stabilisation and cell cycle arrest when knocked out, would appear equally essential in pooled genetic screens using cell proliferation as a readout. Thus, MDM2 would not be identified as differentially required between centrosome-containing and acentrosomal cells in these screening strategies [37–39], highlighting the value of the different approaches taken. Other approaches have linked MDM2 to centrosomes and cell cycle arrest. Cells with supernumerary centrosomes due to failed cell division or PLK4-induced centriole overduplication also undergo p53-dependent cell cycle arrest. In this process, the centrosome activates the **caspase-2/PIDDosome** complex which cleaves MDM2 resulting in p53 stabilisation and p21-dependent cell cycle arrest [44–47]. Since supernumerary centrosomes can lead to extended mitosis, this may provide a further mechanism linked to the mitotic timer response.

While clearly being important for p53-dependent G1 arrest after delayed mitosis, the precise molecular functions of 53BP1 and USP28 required to promote G1 arrest are not fully understood. Initially they were proposed to form a complex with p53 during mitotic delays, with USP28 acting to deubiquitinate and thus stabilise p53 [37,38]. Formation of this ‘stopwatch complex’ was subsequently found to be regulated by **polo-like kinase 1 (PLK1)** in a process integrating mitotic delays over multiple cell cycles [48]. PLK1 is activated in late **Gap2 phase (G2)**, and goes on to orchestrate many conserved functions during mitotic progression and mitotic exit [49]. Timely deployment of PLK1 at centrosomes and kinetochores is tightly regulated during normal mitosis, and it is therefore intriguing that kinetochore localisation control of 53BP1 has been suggested to be under the influence of PLK1 [50]. The activity of this 53BP1- and USP28-containing stopwatch complex has been linked to the DNA damage phosphatase **WIP1** or **protein phosphatase magnesium-dependent 1D (PPM1D)** [48]. Cancer-associated gain-of-function mutations in WIP1 suppress cell cycle arrest after mitotic delays, whereas WIP1 inhibitors overcome this effect [48]. Since the mitotic timer pathway is independent of DNA damage and does not require the DNA damage activated kinases that WIP1 normally counteracts, it is tempting to speculate that WIP1 antagonises PLK1 phosphorylation of stopwatch complexes under these conditions. However, further work is needed to clarify these important questions.

Adding further complexity to this model are the known roles of USP28 and 53BP1 in other p53-dependent processes, including the responses to DNA damage and chemically induced p53 stabilisation with the MDM2 inhibitor Nutlin-3A [51,52]. Both 53BP1 and USP28 were identified in a **short-hairpin RNA (shRNA)** screen looking for targets that, when depleted, allow proliferation in the presence of Nutlin-3A [53], suggesting that they are general factors required for p53-dependent cell cycle arrest. Whether USP28 acts directly on p53 to counteract MDM2-mediated ubiquitination remains unclear, since p53 half-life has been reported to be unaltered in USP28 knockout cells [38]. However, it is important to note that ubiquitin has important signalling and regulatory functions in addition to promoting protein destruction [54]. In agreement with the possibility that p53 activity rather than stability is regulated by stopwatch complexes, one study has demonstrated that both USP28 and 53BP1 knockout cells exhibit reduced p53 occupancy at the

promoters of target genes including p21 [51]. Thus, further work is needed to determine whether USP28 regulates p53 stability or activity, when in the cell cycle this regulation would occur, and whether it acts on additional targets required for G1 arrest.

A model for the mitotic timer

It is clearly remarkable that multiple independent approaches to explain the mitotic timer pathway have converged on a restricted set of positive and negative regulators of p53. However, how these activities function together remains unclear, and several different possibilities exist. In possibly the simplest model, as MDM2 approaches a limiting concentration after mitotic delays USP28–53BP1 could help set the timer threshold by directly antagonising residual MDM2-dependent p53 ubiquitination activity, thus stabilising p53 in early G1 cells. Thus, the impact of falling MDM2 concentration as it approaches a critical threshold would be reinforced if 53BP1–USP28 complexes deubiquitinate p53, stabilising p53 in early G1 (Figure 2B,C). However, the limited effect of USP28 knockout on p53 half-life does challenge this simple model [38]. USP28–53BP1 complex has been shown to control p53 activity rather than directly controlling its stability [51], and could thus act in the G1 daughter cells following a mitotic delay to promote p53-dependent transcription of cell cycle regulators such as p21. In such a scenario, MDM2 could prevent the formation of p53 ‘stopwatch complexes’ during normal mitosis, explaining why these form with apparently slower kinetics than that of MDM2 destruction [11,48]. For cells with supernumerary centrosomes undergoing delayed mitosis, the caspase-2–PIDDosome complex may support G1 arrest by cleaving newly synthesised MDM2, hence further stabilising p53 over a longer timescale [44–46]. Thus, while the core time-sensing mechanism has now been unearthed, how these different activities cooperate remains unclear, and more work is required to dissect how these components synergise to promote p53-dependent transcriptional activity in G1 and exit from the cell cycle.

Mitotic surveillance pathways in disease: protectors or problems?

The SAC is often viewed as the crucial mechanism ensuring the fidelity of chromosome segregation [1]. Indeed, mutations in conserved SAC genes were initially proposed to be drivers of genome instability in cancers [55]. However, as more evidence emerged it became clear that the SAC is important for the survival of aneuploid tumour cells [22] and is thus infrequently lost or mutated in cancer [56]. By contrast, p53, its regulators, and its downstream effectors are mutated with high frequency in the majority of aneuploid human cancers [56]. The p53-dependence of the mitotic timer therefore implies that p53-pathway-deficient cancer cells would be insensitive to mitotic delays and would continue proliferating even following prolonged mitosis. Indeed, this has been shown in cell culture where inactivation of p53 or p21 abolishes the mitotic timer response [8,9,11,37–39,48]. Adding further complexity, cells which retain wild-type p53 functionality can still suppress the mitotic timer response through MDM2 amplification [11]. This is clinically relevant as MDM2 is known to be amplified in a range of p53 wild-type human cancers [56–59]. For cells entering mitosis with higher levels of MDM2, the time in mitosis threshold required for G1 arrest is greatly extended [11], suggesting that MDM2 amplified tumours tolerate longer mitotic delays before arresting but are not completely defective for the pathway. It is within this context that the SAC becomes tumour-sustaining, rather than protective for the organism. Without a functional mitotic timer, SAC-dependent mitotic arrest simply allows more time for cancer cells to align their often aneuploid chromosome complement. Thus, in terms of protecting the organism, the critical failsafe of the SAC is the MDM2–p53 mitotic timer pathway.

Cancer is not the only disease state potentially associated with p53 and the mitotic timer response. In mouse models, p53 stabilisation due either to the introduction of specific point mutations or through the expression of hypomorphic MDM2 alleles creates a range of specific

developmental disorders, many of which have similarities to human diseases [3,60]. Remarkably, knockout of p53 can rescue mouse models for a variety of human developmental disorders (such as microcephaly), suggesting that reduced cell proliferation triggered by p53 activation rather than cell level defects may be the primary cause of these disease states [3,60]. Furthermore, centrosome defects that delay mitosis create microcephaly phenotypes which are rescued by additional knockout of p53, 53BP1, or USP28 [61]. Importantly, such microcephalic phenotypes cannot be rescued by deletion of the DNA damage signalling kinases **ataxia-telangiectasia mutated (ATM)** or **Checkpoint kinase 2 (Chk2)** [62], suggesting that reduced brain size upon centrosome loss may be due to mitotic timer-mediated channelling of cells into a G1 arrest, and not DNA damage signalling. Rather than mounting a sustained G1 arrest, cells were found to undergo arrest followed by cell death [61]. This may represent a key difference between timer activation *in vitro* and *in vivo*. Indeed, p53 is also a known regulator of apoptosis, and p53-dependent apoptosis was reported in neural progenitor cells that experienced a prolonged mitosis [63–65]. That said, cells in culture lack immune surveillance, so although timer-arrested cells appear to mount a robust and sustained G1 arrest under these *in vitro* conditions, it remains possible and even likely that these arrested cells would be cleared by the immune system *in vivo* [12]. Clearly, more research aimed at investigating the fates of timer-arrested cells in different models will provide insights into how, if, and when these arrested cells are removed from the population.

Concluding remarks

A fundamental property of cell cycle checkpoint pathways is their ability to prevent cell cycle progression and enable protective responses such as the repair of DNA damage or correction of error states during chromosome alignment in mitosis. However, extended periods of checkpoint activation may indicate sustained stress or irreparable damage, necessitating additional responses beyond those manifesting within the initial delay. We have discussed herein the concept that time is a danger signal integrating spindle checkpoint signalling in mitosis with the G1 MDM2 and p53-dependent checkpoint pathway. In this mitotic timer response, MDM2 stability sets an upper limit on the time of SAC activation, beyond which cells are triaged out of the proliferative cell cycle. Although considerable progress has been made in identifying key components of the mitotic timer pathway, many questions about this mechanism remain to be addressed (see [Outstanding questions](#)).

Although aneuploid cells can show greater dependency on the SAC, the relationship between aneuploidy and mitotic delays still requires further study. Because aneuploid cancer cell lines are typically deficient for the p53 pathway, they cannot respond to mitotic delays via the timer pathway and often undergo mitotic cell death or collapse from mitosis to G1 without dividing, a phenomenon sometimes referred to as mitotic slippage [66]. Study of the effects of aneuploidy on the mitotic timer are further complicated since p53 wild-type cells that fail mitosis or cytokinesis enter G1 with supernumerary centrosomes and undergo p53-dependent G1 arrest in a tetraploid state [45]. Therefore, it is challenging to definitively determine the effects of whole-genome duplication on the mitotic timer response and to determine whether cell cycle arrest is due to mitotic delays, secondary consequences related to supernumerary centrosomes, or a combination of both. Thus, further studies are required to understand how ploidy changes in p53 wild-type cells affect mitotic timings and subsequent responses in G1. A related question concerns the consequences of mitotic timer pathway activation *in vivo*, and to what extent it explains the presence of senescent cells in tissues. Whether timer-arrested cells would persist *in vivo* or be cleared by the immune system is unknown, although p53 is known to induce the expression of target genes involved in immune signalling and apoptotic responses [67]. Such insights may be valuable for informing therapeutic strategies in cancer which use antimitotic drugs, since these will activate the mitotic timer pathway in p53 wild-type normal cells but not p53-mutant tumour cells which will continue to proliferate and possibly become sensitised to drugs inducing genotoxic stress in

Outstanding questions

Once cells are channelled into a G1 arrest following mitotic delays, are they ever able to exit and enter the next cell cycle? Determining whether cells are quiescent and retain capacity to proliferate, or senescent and destined to be trapped in **G0**, would go some way to addressing this. Cells in culture are not under surveillance by an immune system, so it is formally possible that immune clearance is the physiological response to deal with these arrested, problematic cells. Clearly, timer-arrested cells mount a robust, sustained G1 arrest, but the limits of arrest duration and fate of these cells both in culture and *in vivo* remain to be determined.

While p53-dependent synthesis of p21 maintains the G1 arrest, other target genes (including MDM2) are also up-regulated. How then is p53 able to tolerate increasing MDM2 in G1 and maintain transcriptional capacity to enforce the G1 arrest? More mechanistic work is needed in this area to decipher the molecular underpinnings maintaining the G1 arrest.

It has been proposed that successive delayed but sub-threshold mitotic divisions can lead to p53-dependent cell cycle arrest, implying that there is a memory in the system. Yet a single mitotic delay is sufficient to trigger robust and sustained G1 arrest. Thus, given the apparent binary nature of the mitotic timer, the requirement for memory in the system is not clear.

Although no DNA damage has been detected in the mitotic timer response, many of the p53 pathway components are shared between the timer and DNA damage responses. How DNA damage in the preceding G2 phase would impact the mitotic timer is thus an interesting question. In principle, cells could enter mitosis with both elevated p53 and target genes (including MDM2). How this new steady state would affect the timer response warrants further investigation.

other phases of the cell cycle. Furthermore, while MDM2 inhibitory drugs have not found favour in the clinic in terms of their potential to reactivate p53 in p53 mutant cancers [67], in the specific case of aneuploid MDM2-amplified but p53 wild-type tumours, MDM2 inhibitors or PROTACs could be employed to reduce the threshold for mitotic timer activation, allowing p53 to trigger cell cycle arrest in the ensuing G1, thus suppressing further proliferation.

Finally, whether the concept of time as a danger signal can be extended to other checkpoints, and whether sustained activation of these checkpoints can influence outcomes across distinct cell cycle boundaries are questions worthy of further consideration. An obvious example would be the response to DNA damage in G2 phase prior to mitotic entry [2,68], which would be expected to stabilise p53 and trigger increased expression of p21 and MDM2 [69], with consequences for both cell cycle progression and the mitotic timer pathway (see Outstanding questions). To fully address these complex questions and the nature of time as a danger signal requires quantitative analysis of damage repair processes and different checkpoint signalling pathways, combined with accurate tracking of single cell behaviour across cell cycle phases. Such approaches underpin the exciting and important advances discussed here and have created the framework for future studies on the roles of cell cycle checkpoints in health and disease states.

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Declaration of interests

The authors declare no competing interests.

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